THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The thing that we deliberately change in an experiment is the: 1

(a) conclusion **(b)** unit of measurement
(c) variable we control **(d)** result we observe

2. Earth lies at a distance from the Sun where water: 1

(a) is always steam **(b)** is always ice
(c) is absent **(d)** can remain liquid

3. The gas released by yeast during fermentation of sugar is: 1

(a) hydrogen **(b)** carbon dioxide **(c)** oxygen **(d)** nitrogen

4. Diseases lasting more than three months are called: 1

(a) infectious diseases **(b)** chronic diseases
(c) acute diseases **(d)** deficiency signs

5. Washing hands with soap helps by: 1

(a) producing antibiotics **(b)** adding vitamins
(c) removing pathogens **(d)** giving immunity

6. An electromagnet becomes much stronger when the coil is wound over: 1

(a) a wooden stick **(b)** an iron core **(c)** a glass rod **(d)** a plastic rod

7. In a dry cell, the negative terminal is the: 1

(a) metal cap **(b)** carbon rod **(c)** zinc container **(d)** paste

8. Friction also acts on objects moving through: 1

(a) no medium **(b)** only solids
(c) liquids and gases **(d)** vacuum

9. The base of a dam is broader than its top because: 1

(a) pressure is zero below **(b)** it looks better
(c) water is lighter below **(d)** pressure is largest near the bottom

10. Safe action during lightning is to: 1

(a) use a metal umbrella
(b) stand under a tree
(c) crouch low in an open area away from tall objects
(d) lie flat on the ground

11. Liquids have: 1

(a) fixed shape and volume **(b)** definite volume but no fixed shape
(c) no volume but fixed shape **(d)** neither volume nor shape

12. On heating iron filings and sulfur, the product is: 1

(a) iron oxide **(b)** salt
(c) iron sulfide **(d)** sulfur dioxide only

13. Air is a: 1

(a) mixture of gases **(b)** mineral **(c)** compound **(d)** element

14. In a salt–water solution, the solvent is: 1

(a) salt **(b)** water **(c)** glass **(d)** air

15. Which mirror is used as a rear-view mirror in vehicles? 1

(a) convex mirror **(b)** concave mirror **(c)** lens **(d)** plane mirror

16. A lunar calendar is based on the cycle of: 1

(a) stars only **(b)** Moon's phases **(c)** Earth's seasons **(d)** Sun spots

17. If producers decrease in an ecosystem, the number of herbivores will probably: 1

(a) stay constant **(b)** decrease **(c)** increase **(d)** double

18. Which is an abiotic component? 1

(a) soil **(b)** grass **(c)** tiger **(d)** fungus

19. The biosphere consists of: 1

(a) life **(b)** air **(c)** water **(d)** rocks

20. In birds and mammals, fertilisation takes place: 1

(a) in water **(b)** inside the female body
(c) in nests only **(d)** on land outside

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What is Spirulina and why is it useful? 2

22. Why is prevention better than cure? 2

23. Why is a dry cell called a single-use cell? 2

24. What is a sea breeze? 2

25. Name the gas formed when iron reacts with dilute hydrochloric acid and write the equation. 2

26. Why does ice float on water? 2

27. Define commensalism with an example. 2

28. Why is reproduction essential? 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Give three examples of investigations a student can do at home. 3

30. State any three differences between mass and weight. 3

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31. Describe what would happen to a chair if all its constituent particles were removed. 3

32. Why are ENT specialists' mirrors concave? 3

33. How are seasons related to Earth's motion? 3

34. Why does sexual reproduction help living things survive better? 3

35. Why does a bulb glow with a battery of two cells brighter than one cell? 3

OR

What are the two ways to reduce the heating effect in household circuits? 3

36. How do human activities threaten ecosystems? Give three examples. 3

OR

Explain the role of national parks and sanctuaries. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. Discuss the symptoms, causes and prevention of tuberculosis and hepatitis A. 5

OR

Explain why diseases such as smallpox have been eradicated but others like influenza return every year. 5

38. Write an account of how air pressure and wind are connected, including breeze, storm and cyclone. 5

OR

A tank 3 m high is on a roof. Explain why the top floor gets a weak stream. Also calculate pressure when a force of 900 N acts on 0.3 m². 5

39. Describe the use of the measuring cylinder and balance with precautions to read the meniscus. 5

OR

Write about solid–liquid, liquid–liquid, gas–liquid and solid–solid solutions with examples. 5

40. Draw and explain a grassland food web containing grass, grasshopper, mouse, frog, snake and hawk, and say what happens if snakes are removed. 5

OR

Explain ecosystem services and why conservation is necessary. 5

END OF QUESTION PAPER — Total: 80 Marks

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